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Characterisation of Redlen high-flux CdZnTe

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ABSTRACT: CdZnTe is a promising material for the current generation of free electron laser light sources and future laser-driven γ -ray sources which require detectors capable of high flux imaging at X-ray and γ -ray energies (> 10 keV). However, at high fluxes CdZnTe has been shown to polarise due to hole trapping, leading to poor performance. Novel Redlen CdZnTe material with improved hole transport properties has been designed for high flux applications. Small pixel CdZnTe detectors were fabricated by Redlen Technologies and flip-chip bonded to PIXIE ASICs. An XIA Digital Gamma Finder PIXIE-16 system was used to digitise each of the nine analogue signals with a timing resolution of 10 ns. Pulse shape analysis was used to extract the rise times and amplitude of signals. These were measured as a function of applied bias voltage and used to calculate the mobility (μ) and mobility-lifetime ($\mu\tau$) of electrons and holes in the material for three identical detectors. The measured values of the transport properties of electrons in the high-flux-capable material was lower than previously reported for Redlen CdZnTe material ($\mu_e\tau_e \sim 1 \times 10^{-3} \text{ cm}^2\text{V}^{-1}$ and $\mu_e \sim 1000 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$) while the hole transport properties were found to have improved ($\mu_h\tau_h \sim 3 \times 10^{-4} \text{ cm}^2\text{V}^{-1}$ and $\mu_h \sim 100 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$).

KEYWORDS: Charge induction; Materials for solid-state detectors; Gamma detectors (scintillators, CZT, HPG, HgI etc); X-ray detectors

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Contents

1	Introduction	1
2	Theory	2
2.1	Shockley-Ramo theorem	2
2.2	Small pixel Hecht equation	2
2.3	Mobility Calculation	3
3	Experimental Details	4
4	Pulse processing techniques	5
5	Results	6
5.1	Electron mobility-lifetime product	6
5.2	Electron mobility	8
5.3	Hole mobility-lifetime product	9
5.4	Hole mobility	10
6	Conclusion and discussion	11

1 Introduction

CdZnTe has become the semiconductor of choice as a direct conversion detection material for high energy X-rays and γ -rays, due to a wide bandgap of 1.44 eV at 300 K and excellent electron transport properties, leading to a low leakage current and high spectroscopic resolution at room temperature. A high effective atomic mass ($Z_{\text{eff}} \sim 50$) and density of 5.8 g cm^{-3} provides good stopping power for high energy photons, making CdZnTe the current first choice for pixelated x-ray and gamma-ray detectors at room temperature [1].

High-flux detection is desirable for use in several applications such as imaging at next generation free electron light sources and medical imaging, in which detection of a photon flux of $> 10^8 \text{ photons cm}^{-2}\text{s}^{-1}$ is required. The flux that can be achieved with standard CdZnTe is currently limited to $\sim$ a few Mcps mm^{-2} [2] by polarisation, in which holes trapped by defects generate space charge, resulting in a non-uniform electric field [3]. Therefore, one method for improving detection at high X-ray fluxes requires the development of CdZnTe with better hole transport properties (e.g. charge carrier mobility and lifetime), in which hole trapping is reduced.

Recently, Redlen Technologies have produced an alternate growth and fabrication process that allows production of CdZnTe with improved hole transport properties and reduced trapping optimised for high flux applications. In order to characterise the charge transport properties of this material, small pixel CdZnTe devices were flip-chip bonded to the PIXIE ASIC, which has

been developed at STFC. [4] Pulse shape analysis was used to perform rise time and amplitude measurements as a function of applied bias. These measurements were used to extract the charge transport properties of the material, which are essential for understanding the performance and suitability of this CdZnTe for future high-flux applications.

2 Theory

2.1 Shockley-Ramo theorem

The Shockley-Ramo theorem describes the process of charge induction in a detector due to the motion of charge carriers using the concept of a weighting potential. The weighting potential, $\varphi_0(x)$, is derived by solving the Laplace equation for the conditions where the electrode of interest is at unity and the remaining electrodes are held at zero potential for no trapped charges [5, 6].

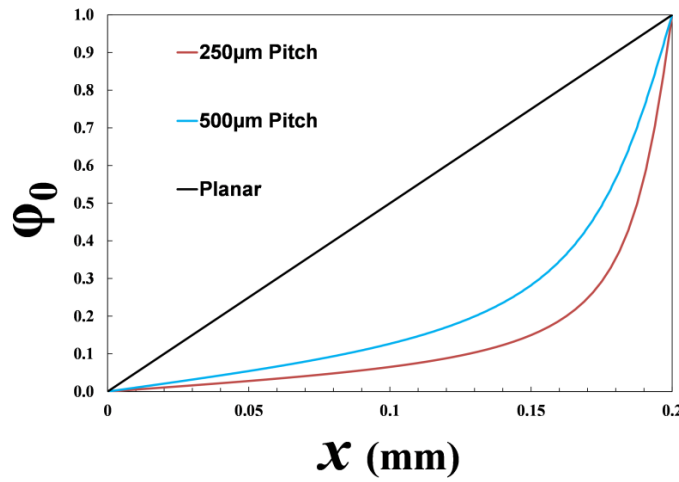


Figure 1. Weighting potentials for small pixels at different pitches, with a detector thickness of 2 mm.

In the simplest geometry with planar electrodes, the weighting potential rises linearly from cathode to anode. In a detector in which the pixels are small relative to the thickness of the detector material, the weighting potential becomes constrained close to the pixel. Weighting potentials for a 2 mm thick detector with different pixel pitches, shown in figure 1, were calculated using Sentaurus TCAD, a branch of electronic design software used to model the operation of semiconductor devices [7]. The induced charge on an electrode is proportional to the change in weighting potential that a charge experiences as it moves through the detector. If a negative bias voltage is applied to the large planar electrode then the signal in the detector pixels will be dominated by the transport of electrons. If the bias voltage is positive then it will be the transport of holes that dominates the signal. This is known as the small pixel effect [8].

2.2 Small pixel Hecht equation

The ratio of charge created by an event in the detector material to the charge collected at the electrical contacts is known as the charge collection efficiency (CCE). Due to trapping of charge carriers, the CCE is a function of the distance travelled by the charges and the drift time.

The Single Carrier Hecht Equation for a planar detector [9], as given in equation (2.1), gives the CCE as a function of bias voltage, making the assumptions of no de-trapping and a uniform electric field. In order to use the single carrier version of this equation, an assumption of a small interaction depth must also be made. For a photon energy of 60 keV the mean free path in CdZnTe is of the order 0.28 mm. This was calculated using XCOM data [10]. When compared to a detector thickness of 2.00 mm it is reasonable to assume that the interaction depth is negligible and the transport will be dominated by a single carrier type.

$$CCE = \frac{Q}{q_0} = \left(\frac{\mu\tau \cdot V}{d^2} \right) \left[1 - \exp\left(-\frac{d^2}{\mu\tau \cdot V} \right) \right] \quad (2.1)$$

Here, q_0 is the initial charge created by the X-ray, Q is the charge induced on the electrical contact (i.e. the amplitude of the charge pulse), $\mu\tau$ is the charge carrier mobility-lifetime product, V is the applied bias and d is the detector thickness. The Hecht relation can be used to study the charge transport properties of both electrons and holes. In this experiment, measurement of the properties of both charge carriers was achieved by switching the bias polarity.

The Single Carrier Hecht Equation shown in equation (2.1) was derived for a planar detector in which the weighting potential is linear, as shown in figure 1. Carriers drifting in areas of the detector where the weighting potential is diffuse will only induce a small charge on each electrode. Charges trapped in these areas of the detector will result in a charge loss that appears greater than in the simple planar detector, which is unaccounted for by equation (2.1).

To account for the non-linear weighting potential, which leads to an additional depth dependency of the CCE, a modified version of the equation was used [11].

$$CCE = \frac{Q}{q_0} = N \cdot \left[\left(\frac{d}{\mu\tau \cdot V} \right) \cdot \int_0^d \varphi_0(x) \cdot e^{-\frac{d-x}{\mu\tau \cdot V}} \cdot dx + e^{-\frac{d^2}{\mu\tau \cdot V}} \right] \quad (2.2)$$

Here $\varphi_0(x)$ is the weighting potential introduced in section 2.1, x is the interaction position and N is a correction coefficient, to correct for errors in the energy calibration or to act as a levelling constant when no calibration was used. In order to extract $\mu\tau$ for electrons and holes from experimental data, equation (2.2) was solved numerically using the simulation results shown previously in figure 1.

A comparison of the two models with $\mu\tau = 1 \times 10^{-4} \text{ cm}^2\text{V}^{-1}$ is presented in figure 2, which demonstrates the reduced CCE at lower biases. As seen in figure 2, the curve predicted by the modified Hecht equation is significantly different to that of the standard Hecht at voltages $< 50 \text{ V}$; this is due to the majority of carriers becoming trapped before drifting in areas of the detector where the weighting potential has a significant magnitude.

2.3 Mobility Calculation

By measuring the rise time of charge pulses created by X-ray interactions it is also possible to extract the charge carrier mobility for electrons and holes using equation (2.3).

$$\mu = \frac{v}{E} = \frac{d'}{t} \cdot \frac{T}{V} \quad (2.3)$$

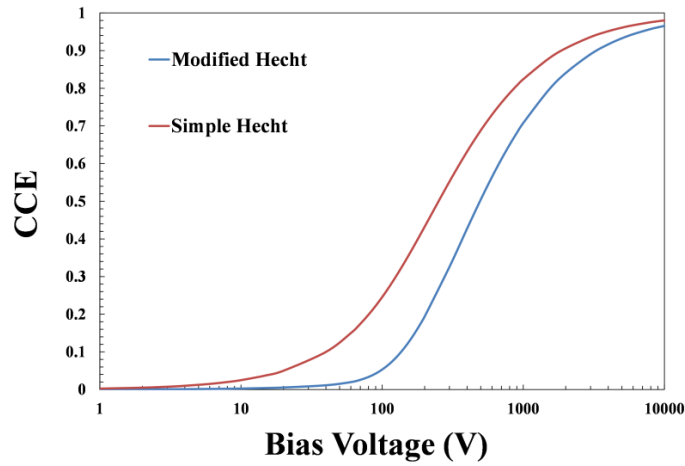


Figure 2. A plot of equation (2.1) and equation (2.2) using $\mu\tau = 1 \times 10^{-4} \text{ cm}^2\text{V}^{-1}$, to demonstrate the differences between the models.

Here, v represents the charge carrier velocity, E is the electric field strength applied across the detector, μ is the charge carrier mobility, T is the thickness, d' is the effective distance over which the signal is induced and t is the rise time.

In small pixel detectors, where the weighting potential is non-linear, charge is only induced when carriers drift in regions of the detector where the change in weighting potential has a significant value. The effective distance is used to account for the drift length over which the detector signal is induced. Details of the calculation of the effective distance are given in section 5.2. An assumption of a uniform electric field is also made.

3 Experimental Details

An Application Specific Integrated Circuit (ASIC), known as the PIXIE ASIC, was developed at STFC RAL to study charge induction effects in pixelated detectors. [4] The ASIC consists of four arrays of nine channels, three with pitches of $250 \mu\text{m}$ and one with $500 \mu\text{m}$. Each pixel contains a charge sensitive pre-amplifier, output buffers and leakage current compensation circuits designed to provide immunity to leakage current up to 250 pA . The pixels do not contain any shaping circuits — instead, the analogue waveform is sampled directly from the chip. The ASIC is also able to operate in two ranges, high gain (for measurements up to 150 keV) and low gain (for measurements up to 1.5 MeV). In high gain mode the ASIC is limited to a leakage current of $\sim 250 \text{ pA}$ which, for the current detectors, limited the ASIC operation to the low gain mode [4].

Whilst only one electrode on the CdZnTe detector was pixelated, by switching the polarity of the applied bias, both holes and electrons were detected. While the PIXIE ASIC can measure both positive and negative polarity signals, the leakage current compensation circuit has been optimised for the detection of electrons. This limited the measurement of hole signals to smaller biases than were used to study electrons.

Only one of the 3×3 arrays is read out at any one time and this was selected using a 2-bit input to the ASIC. The output from each of the pixels of the selected array was digitised using an XIA

DGF PIXIE-16 system, performing 16-bit digitisation at a maximum rate of 100 MHz. The digitiser was operated in list mode to allow simultaneous capture of pulse shapes from all nine pixels, with timing resolution of 10 ns and a pulse length of $2\ \mu\text{s}$ and $7\ \mu\text{s}$ for electrons and holes respectively, with the longer pulse shapes for holes used to account for the slower rise times of these signals.

The CdZnTe and electrical contacts were fabricated by Redlen Technologies from high-flux-capable material using a proprietary method. Detectors have dimensions of $4.9\text{ mm} \times 4.9\text{ mm} \times 2\text{ mm}$ were flip-chip bonded to the PIXIE ASIC using a gold stud and low-temperature-curing silver epoxy technique developed at the STFC RAL [12]. During operation, the detector was placed into an enclosure to allow cooling in a dehumidified environment. The temperature and humidity in the enclosure were controlled using a Peltier cooler, chiller and dehumidifier. Measurements were taken at a temperature of $303 \pm 1\text{ K}$, the intended operating point of these detectors, and the relative humidity was kept below 10%. A bias was applied across the detector bulk using a Keithley 2410-C, which was also used to measure current drawn by the entire detector. Data was recorded by uniformly irradiating the detector with an Am-241 source of activity 185 MBq at a distance of 100 mm. This analysis was repeated for three identically bonded CdZnTe crystals and averaged to account for variations between individual crystals.

4 Pulse processing techniques

The PIXIE-16 system recorded the digitised pulse shape data in 16-bit binary data files. Data was recorded with detection thresholds of 22.0 ADU (Analogue-to-Digital Units) ($\sim 2.8\text{ keV}$) and 40.0 ADU ($\sim 5.6\text{ keV}$) for electrons and holes respectively; these were implemented in the PIXIE-16 control software and were used to distinguish real events from low energy noise introduced by the electronics and cross-talk between pixels.

Charge sharing (CS) in which the charge generated from one X-ray interaction is collected on multiple pixels due to charge carrier diffusion is undesirable as it incorrectly introduces low energy events in multiple pixels. In order to accurately account for CS, only events where the pulse amplitude in the central pixel exceeds the set threshold were recorded. As the central pixel is fully surrounded by neighbouring pixels, CS events were accurately identified from use of coincidence timing. Any frames of data with two or more pixels over the set threshold were not included in the data analysis.

Bespoke code was written in Matlab to extract key characteristics of each of the recorded pulse shapes which are extracted from the binary output of the digitiser. Once extracted, pulse shapes were smoothed with a first order Savitsky-Golay filter (SGF) using a frame length of 15 [13] as shown in figure 3a. This filtering reduced the RMS noise by 4 ADU (0.56 keV). An interpolation of 10 points between each data point was also implemented to improve the calculation of pulse rise times.

The amplitude of each smoothed pulse was extracted and binned into pulse height spectra with a bin size of 0.5 keV. Characteristic peak positions at 59.54 keV (Am-241 γ -ray), 17.72 keV (L_β Neptunium peak) and 13.95 keV (L_α Neptunium peak) were identified from the electron pulse height spectra and centroids calculated using a Gaussian fit to the spectral peaks. A linear fit was used to convert from units of ADU to keV as shown in figure 3b. This gave an energy calibration of $\sim 0.14\text{ keV}$ per ADU. Values of the FWHM of the 59.54 keV peak were $\sim 1.8\text{ keV}$ for both pitches at -1000 V , as previously reported [4].

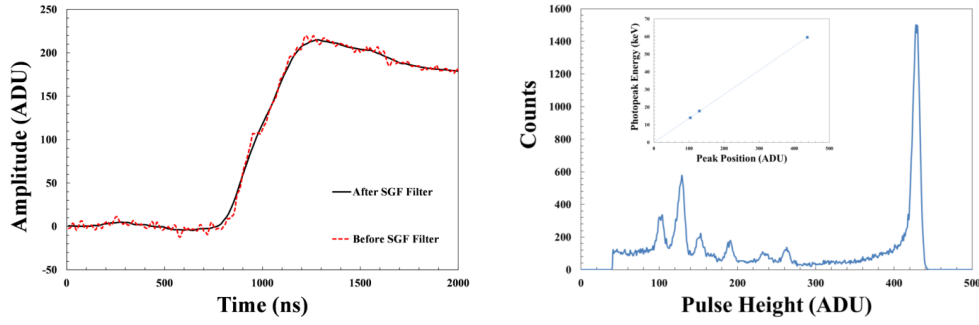


Figure 3. a (Left) — A typical pulse shape before and after SGF filtering. b (Right) — A typical electron calibration spectra for an individual pixel, inlayed with a linear fit to known energies, allowing conversion between ADU and keV.

The calibration calculated from the electron data was not valid for holes as the preamplifier response varied for positive and negative charges. Calibration using hole measurements was not possible due to the poor spectroscopic performance of the detector at high positive biases, which was caused by increased leakage current at high biases. However, by using N in equation (2.3) as an estimate of the energy at 100% CCE (in ADU), it was still possible to extract $\mu_h \tau_h$.

A low energy threshold of 2.8 keV was used to determine whether a signal in a pixel was real or simply noise. A separate Matlab script was used to re-bin pulse height spectra into energy using the calibration data using a 1 keV bin width. The Am-241 59.54 keV peak was fitted with Gaussian distributions to extract the full width at half maximum (FWHM) and centroid value as a function of bias voltage. The centroid at -1000 V was assumed to be 100% CCE for electrons. The measured peak centroids were then converted to CCE relative to the -1000 V centroid. This was repeated for voltages between -1000 V and -100 V and plotted as a function of applied bias. Centroids were measured between 100 V to 400 V for holes and plotted against applied bias, in ADU. The smaller range for holes was due to an increased leakage current at higher applied biases. These were then fitted using the modified Hecht equation (equation (2.3), see section 2.2) and the mobility-lifetime was extracted.

Additionally, the rise time of each event was measured. This was performed by finding the first point at which 20% and 90% of the maximum amplitude was reached, measuring the time between them and linearly extrapolating for the 100% rise time. The 20% threshold was chosen to avoid system noise affecting results. Pulse rise times were compiled into histograms with a bin width of 12.5 ns whilst rejecting all CS events. The rise time histograms were fitted with Gaussian distributions to extract the full width at half maximum (FWHM) and centroid value. The centroid rise time value was taken to be the mean charge carrier rise time. Plots of rise time as a function of applied bias were fitted with equation (2.3) to extract the charge carrier mobility. This analysis was repeated for arrays with pitches of 250 μm and 500 μm .

5 Results

5.1 Electron mobility-lifetime product

In order to calculate the electron mobility-lifetime ($\mu_e \tau_e$), the amplitude of pulse shapes in the range -100 V to -1000 V were measured. Figure 4a shows typical electron pulse shapes at different biases

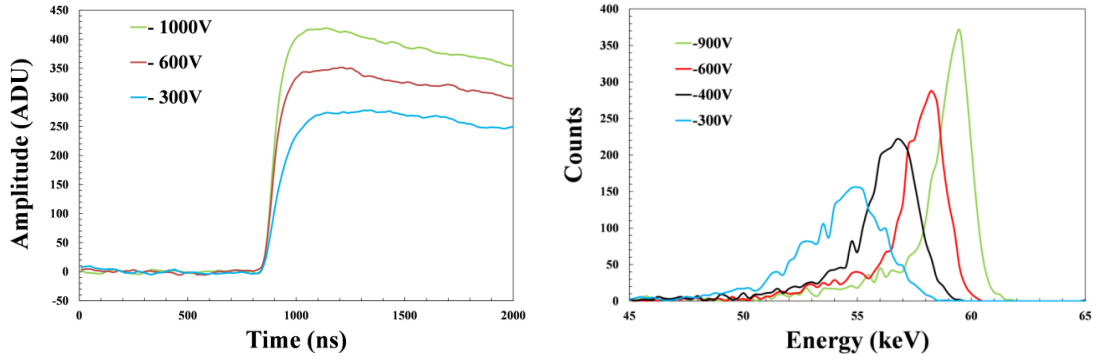


Figure 4. a (Left) — Typical electron pulse shapes, showing the change of amplitude and rise time as a function of bias voltage. The voltages given in the legend are negative. b (Right) — Example of electron energy spectra produced using Redlen CdZnTe and PIXIE ASIC under varying applied biases. Data for both was taken at 303 K on a 500 μm pitch detector.

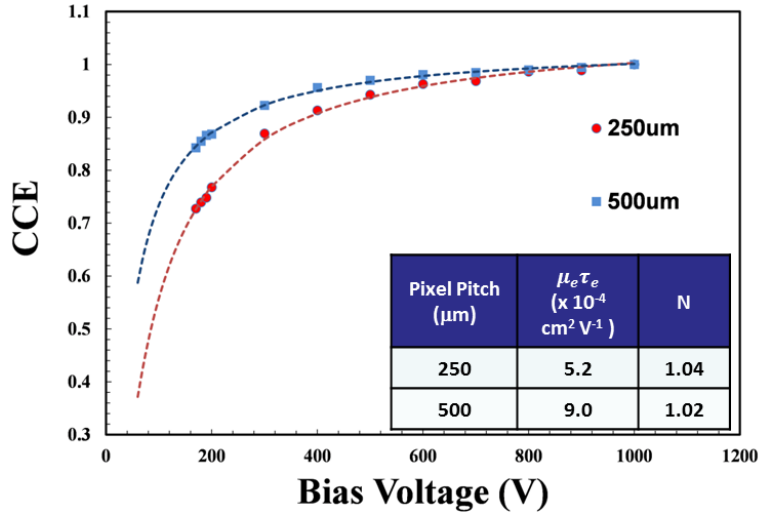


Figure 5. Small pixel Hecht fit for varying bias voltages on high-flux CdZnTe. A multiplicative factor, N , was used to shift the fit up, to account for the CCE at -1000 V not being precisely 1.0.

and the amplitude increasing as a function of bias. Figure 4b shows histograms of the measured pulse heights, calibrated in terms of energy, at different voltages. These histograms show the reduction in the centroid energy and an increasing width of the photo-peak at lower biases. A plot of CCE against bias was fitted with equation (2.3) and $\mu_e \tau_e$ extracted — this is shown in figure 5.

From these fits, values of $\mu_e \tau_e$ were determined to be $5.2 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1}$ and $9.0 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1}$ for pixel pitches of 250 μm and 500 μm respectively. N values close to 1.0 indicate the calibration from energy to CCE was effective and that only a small levelling coefficient was required. The average across three detectors produced a value of 1.1. Typical literature values of $\mu_e \tau_e$ for standard Redlen CdZnTe are of the order $1.0 \times 10^{-2} \text{ cm}^2 \text{ V}^{-1}$. This demonstrates that there is a reduction of over an order of magnitude in the value of $\mu_e \tau_e$ in high flux Redlen CdZnTe when compared to standard Redlen material.

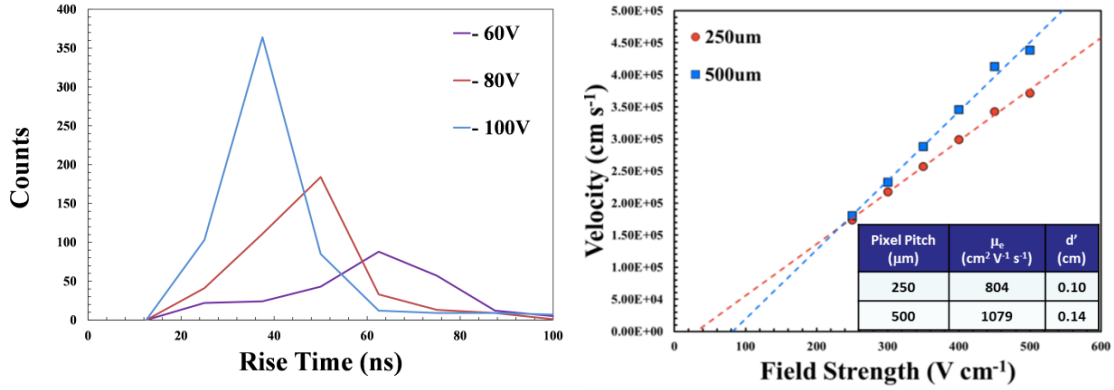


Figure 6. a (Left) — Electron rise time spectra. b (Right) — Electron mobility fits. The gradient of these lines gives the mobility. The effective distance used, d' , is given for each pitch. Both graphs were taken at a temperature of 303 K with the 500 μm pitch array.

The small variation in $\mu_e\tau_e$ between pixel pitches may be explained by several factors including variations in the material properties between arrays due to material non-uniformity, non-uniform electric fields within the material and errors in the weighting potential simulation used to extract $\mu_e\tau_e$ [14, 15].

5.2 Electron mobility

The electron rise times were extracted using the 20%–90% method described previously and had values of the order 50–100 ns. A clear trend of decreasing rise time with increasing bias can be seen in figure 6a. As electron rise times at bias voltages with magnitude greater than -100 V are faster than that of the pre-amplifier, the measurement range was limited between -50 V and -100 V. The effective distance from equation (2.2) accounted for the depth at which the change in weighting potential from a 60 keV event was sufficient to induce a signal on the detector. This was performed by calculating the point at which the change in weighting potential was greater than the 2.8 keV low energy threshold. Values are given in figures 6b and 9b for electrons and holes respectively.

Figure 6b shows a plot of field strength against velocity fitted with equation (2.3). Using the gradient of these lines and the effective thickness, mobility values of $\mu_e = 804 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $\mu_e = 1079 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ were determined for pitches of 250 μm and 500 μm respectively. Similarly to section 5.1, the discrepancy between pitches could be explained by non-uniformities in the material and simulation errors [14, 15].

These measurements show that the electron mobility is unchanged in the high-flux material, compared to a literature value of $1100 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ [16]. As the previous measurement of $\mu_e\tau_e$ was more than an order of magnitude lower than that of standard Redlen CdZnTe, this indicates an increase in the amount of electron trapping in this material. A calculation of the electron lifetime (τ_e) from the $\mu_e\tau_e$ and μ_e data gives a value of the order of 1 μs compared to 10 μs in the standard CdZnTe.

5.3 Hole mobility-lifetime product

The same analysis as section 5.1 was performed with positive bias voltages in order to determine the mobility-lifetime of holes ($\mu_h \tau_h$). Figure 7a shows typical hole pulse shapes. By comparing figures 4a and 7a, the difference in amplitude and rise time between holes and electrons can be seen. The leakage current compensation circuit behaved differently for positive and negative charges and the increased leakage current with positive applied biases reduced the decay time of the preamplifier. The range over which the compensation circuit is able to operate correctly is therefore limited to 250 pA per pixel, corresponding to a maximum applied bias of 400 V. At biases greater than this, the rapid decay of the preamplifier caused loss of signal. The difference in decay time between electrons and holes can be seen by comparing figures 4a and 7a.

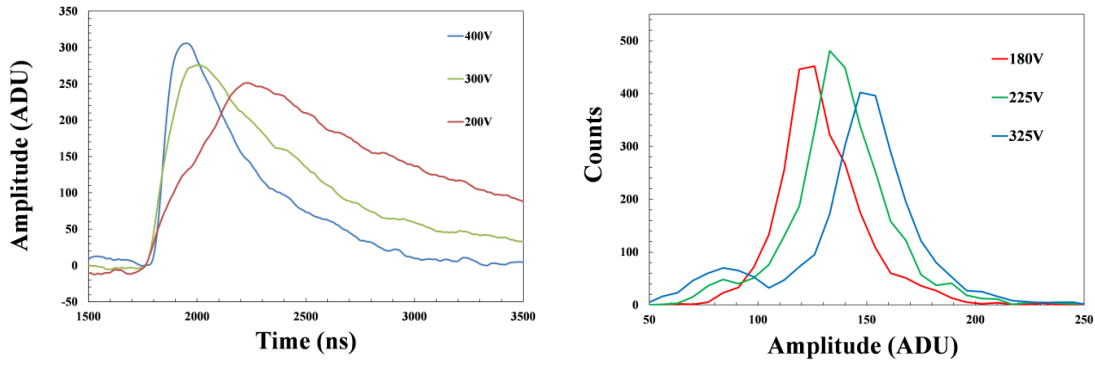


Figure 7. a (Left) — Hole pulse shapes recorded at varying biases. The change in rise time and amplitude as a function of bias should be noted. b (Right) — Hole peak height histogram. The shift in peak position with decreasing applied bias should be noted. Both graphs were taken on the 500 μm pitch array at 303 K.

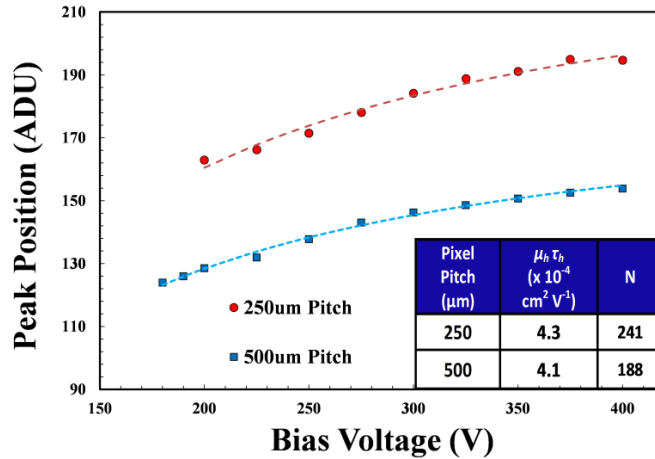


Figure 8. Hole data taken at 303 K in low gain mode. The fits are produced using the small pixel Hecht relation (equation (2.3)) and $\mu_h \tau_h$ values were extracted.

The change in hole pulse height as a function of bias voltage is displayed in figure 7b. Figure 8 shows the peak position as a function of bias. A direct calibration of this data was not possible

due to poor spectroscopic performance at high bias voltages. Therefore, when measuring holes, no calibration was used. In this case, N acted as an estimate of 100% CCE in ADU. $\mu_h \tau_h$ was still consistently extracted from the fit.

Values of $\mu_h \tau_h$ were calculated to be $4.09 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1}$ and $4.26 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1}$ for pixel pitches of $250 \mu\text{m}$ and $500 \mu\text{m}$ respectively. Taking an average across the three detectors produced a value of $(2.9 \pm 1.4) \times 10^{-4} \text{ cm}^2 \text{ V}^{-1}$. Literature sources for $\mu_h \tau_h$ vary, with the highest previous measurement of $2.4 \times 10^{-5} \text{ cm}^2 \text{ V}^{-1}$ in Redlen CdZnTe [14]. The value given for Redlen high-flux-capable CdZnTe was at least an order of magnitude better than the standard Redlen CdZnTe and indicated significantly less hole trapping in the new material. Similarly to section 5.1, the discrepancy between pitches could be explained by non-uniformities in the material and simulation errors [14, 15].

5.4 Hole mobility

A clear trend of decreasing rise time with increasing bias can be seen in the rise time histograms in figure 9a. The rise times for holes were larger than for electrons, meaning it was possible to evaluate mobility data over a larger range of bias voltages down to -1000 V . Figure 9b shows a plot of the measured hole velocity as a function of applied field strength which was fitted with equation (2.3), giving mobility values of $\mu_h = 76 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $\mu_h = 118 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ for pixel pitches of $250 \mu\text{m}$ and $500 \mu\text{m}$ respectively. It also gives the effective distances used, which were calculated in the same way as section 5.2 for the greater low energy threshold. Similarly to section 5.1, the discrepancy between pitches could be explained by non-uniformities in the material and simulation errors [14, 15].

Compared to a literature value of $88 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ [16], these measurements show that the hole mobility in high-flux material was similar to that of standard Redlen CdZnTe. As the measurement of $\mu_h \tau_h$ in the previous section gave a value which was more than an order of magnitude higher than that of standard Redlen CdZnTe, the new material must have had reduced hole trapping. A calculation of the hole lifetime (τ_h) from the $\mu_h \tau_h$ and μ_h measurements gives a value of the order of $3 \mu\text{s}$ compared to $<1 \mu\text{s}$ in the standard CdZnTe.

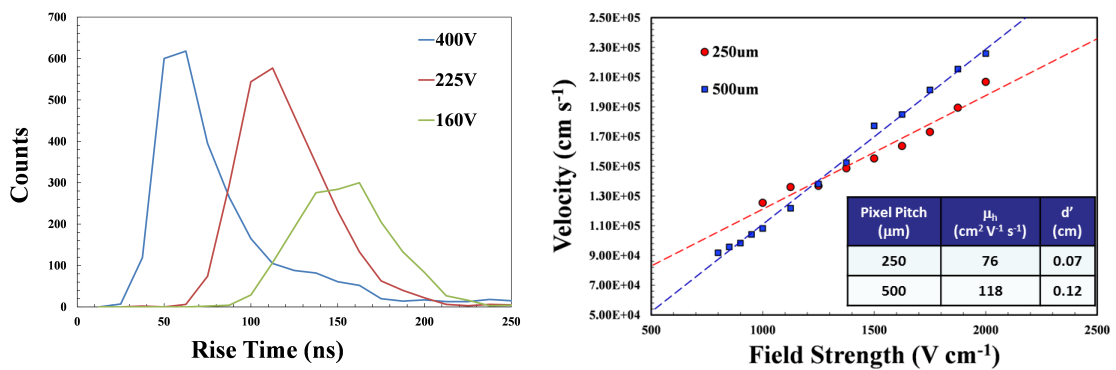


Figure 9. a (Left) — Hole rise time spectra. b (Right) — Hole mobility fits. The gradient of these lines gives the mobility. The effective distance, d' , is given in the inset table. The data for both graphs was taken at a temperature of 303 K with the $500 \mu\text{m}$ pitch array.

6 Conclusion and discussion

Table 1 summarises the results of the charge transport characterisation of three high-flux CdZnTe crystals using these methods. It is clear that the electron charge carrier transport properties in the high-flux material are approximately an order of magnitude poorer than the standard spectroscopic grade Redlen CdZnTe. In comparison, the hole lifetime is over an order of magnitude greater than that previously reported and is now very similar to that of the electrons.

This improvement in the hole lifetime was likely due to a reduction in the concentration of deep hole traps within the material. However, this improvement in hole lifetime is associated with a reduction in the electron lifetime which was likely due to an increased concentration of deep electron traps. For applications like spectroscopy, where achieving the best possible energy resolution is desirable, standard Redlen CdZnTe is the most suitable material due to the better electron charge transport properties. However, in high flux applications, the higher hole lifetime in this new CdZnTe material will provide superior performance due to a reduction in trapping/polarisation. These results are consistent with that of Iniewski et al., which showed that the material performed well at fluxes up to 200 Mcps/mm² for medical imaging applications [2].

Table 1. A summary of the measured charge transport properties of three “high-flux” Redlen CdZnTe detectors [14, 16].

	$\mu_e \tau_e$ ($\times 10^{-4} \text{ cm}^2 \text{ V}^{-1}$)	μ_e ($\text{cm}^2 \text{ V}^{-1} \text{ s}^{-1}$)	τ_e ($\times 10^{-6} \text{ s}$)	$\mu_h \tau_h$ ($\times 10^{-4} \text{ cm}^2 \text{ V}^{-1}$)	μ_h ($\text{cm}^2 \text{ V}^{-1} \text{ s}^{-1}$)	τ_h ($\times 10^{-6} \text{ s}$)
High Flux CdZnTe	11 ± 6	940 ± 190	1.2 ± 0.8	2.9 ± 1.4	114 ± 22	2.5 ± 1.4
Standard CdZnTe	100	1100	11	0.2	88	0.2

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